CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

Memory Organization

Lecture No.- 1

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Recap of Previous Lecture

Topic Direct Memory Access (DMA)

Topic Cycle Stealing

Topic Burst Mode

Topics to be Covered

Topic Memory Hierarchy

Topic Memory Presentation

[NAT]

GATE-PYQ

#Q. A device with data transfer rate 10KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 microsec. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode?

data transfer time = 0

$$= \frac{\text{Programmed I/O time}}{\text{Interrupt I/O time}} = \frac{100\,\mu sec}{4\,\mu sec} = 25$$

for 10 KB, time = 1 sec

for 1 B, time $= \frac{1 \text{ sec}}{10 \text{ K}\cancel{\text{B}}} * 1 \cancel{\text{B}}$

$= 0.1$ msec

$= 100$ µsec

[NAT] GATE-PYQ

#Q. On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

Initialize the address register — 1
Initialize the count to 500 — 1
→ 2

LOOP: Load a byte from device — 2
Store in memory at address given by address register — 2
Increment the address register — 1
Decrement the count — 1
If count !=0 go to LOOP — 1

500 times

7 * 500 = 3500

Total = 3502

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

Interrupt I/O :-

500 B transfer from I/O to mem

DMA :-

20 cycles for initializatⁿ

2 cycles ⟹ for one byte

DMA time = 20 + 500 * 2 = 1020 cycles

$$\text{Speed up} = \frac{\text{Interrupt I/O time}}{\text{DMA I/O time}} = \frac{3502}{1020}$$

$$= 3.43$$

Memory Organization

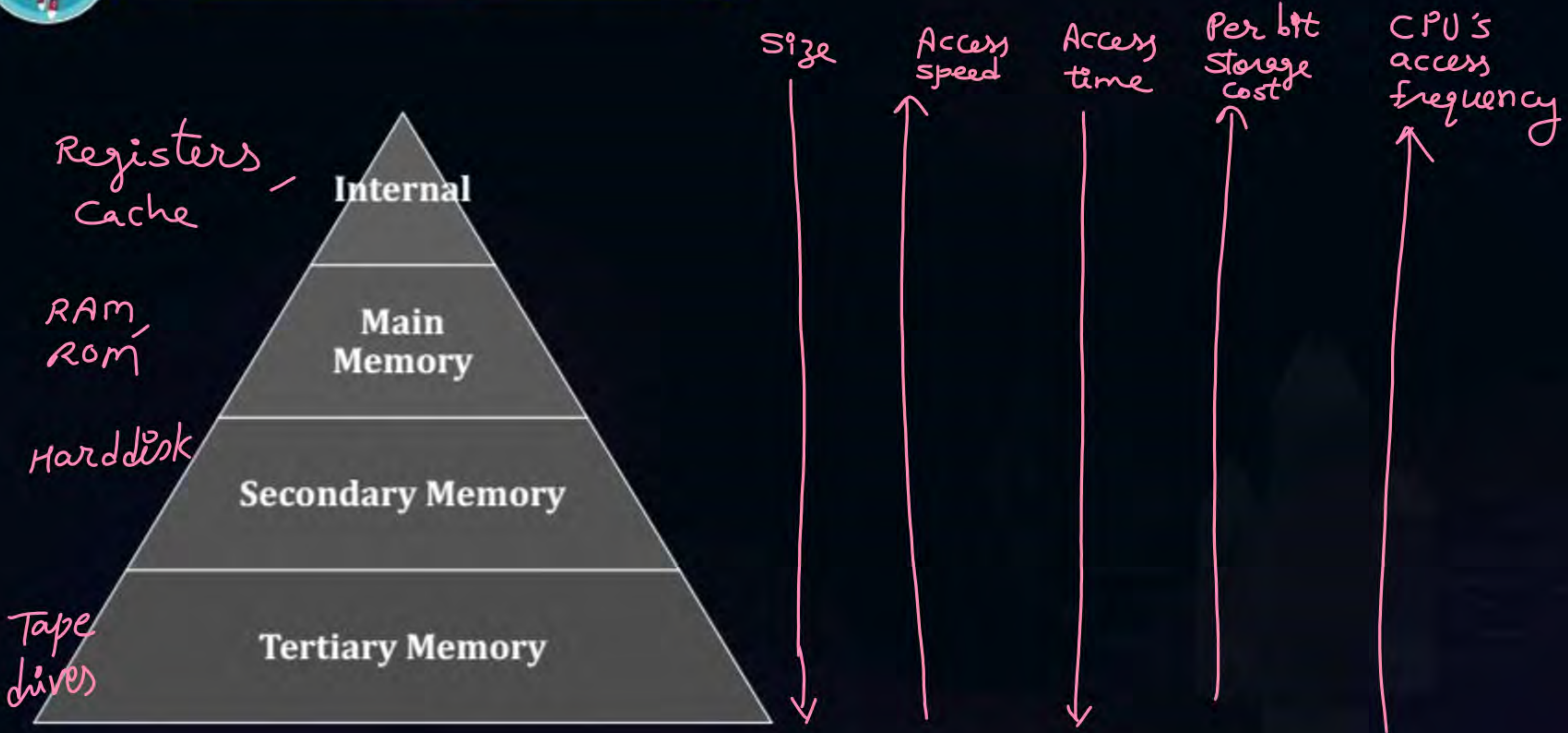
Topic : Memory Hierarchy

Memory hierarchy used when discussing performance issues.

Goal of Memory Hierarchy:

1. To maximize the Access Speed
2. To minimize the Per Bit Storage Cost

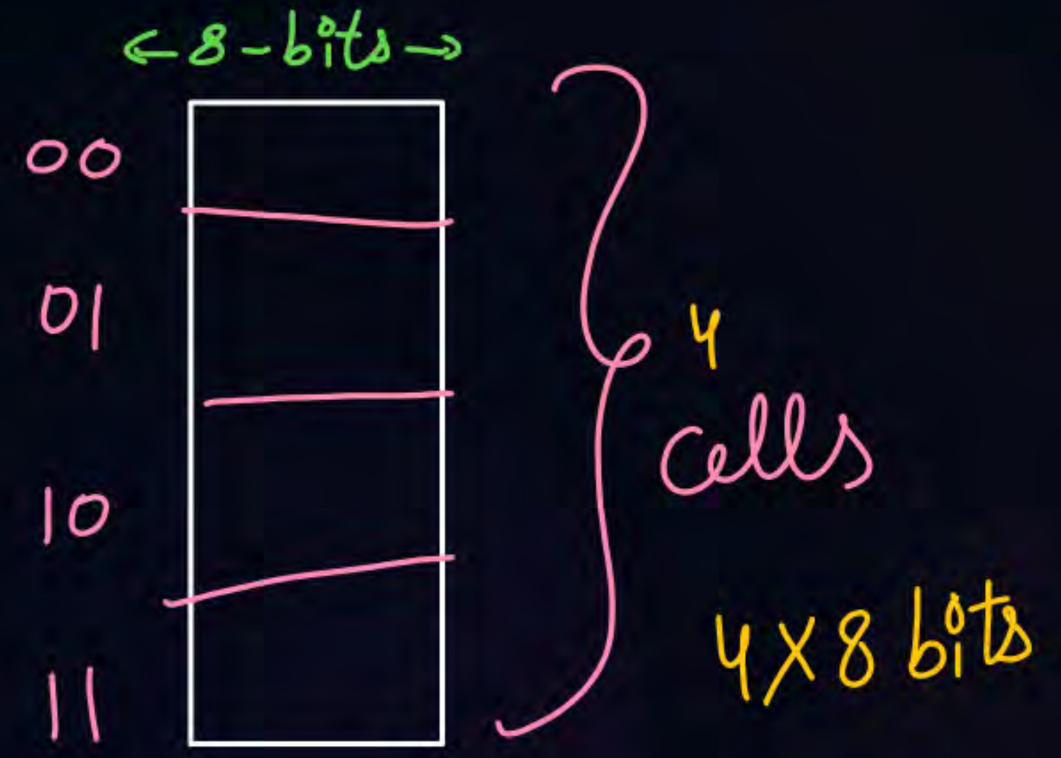
Topic : Memory Hierarchy



Topic : Memory Presentation

$= \text{No. of cells} \times 1 \text{ cell capacity}$

$= \text{No. of memory locations} \times \text{bits per location}$



ex:-

$\underline{128K} \times 16 \text{ bits} \Rightarrow 2^{17} \times 16 \text{ bits}$

or

$2^{17} \times 2 \text{ bytes}$

or

$128K \times 2 \text{ bytes}$

↓

address size = 17 bits

Topic : Memory Presentation

Byte addressable memory:-

ex:- 256K × 8 bits

or

256K × 1 byte

or

256k bytes

[MCQ]

#Q. Memory is represented as?

A $A \times B$ where A = No. of memory locations, B = No. of bits in each location

B $2^a \times B$ where a = No. of address bits, B = No. of bits in each location

C $B \times A$ where, B = No. of bits in each location, A = No. of memory locations

D (A) & (B) both

[MCQ]

#Q. A memory has 14-bits address bus. Then how many memory locations are there?

$2^{14} = 16k = 16384$

A 16K

B 16384

C 2^{14}

D All

Memory cycle:-

Time needed to perform read or write on one address in memory.

ex:- Byte addressable mem.

mem. cycle time = Time to read or write 1 byte from mem.

ex 2:- word addressable mem.

mem. cycle time = Time to read or write 4 bytes content

[MCQ]

#Q. The memory cycle time of a memory is 200nsec. The maximum rate with which the memory can be accessed?

Note: Consider memory as byte addressable.

A) 500 Bytes / Sec

B) 2000 Bytes / Sec

C) ✓ 5 Mbytes / Sec

D) 5 GBytes / Sec

in 200nsec, from memory data = 1 Byte

in 1 sec — 11 —— $= \frac{1\ B}{200 * 10^{-9} sec}$

$= \frac{10^{9}\ B}{200\ sec}$

$= 5\ MB/sec$

[NAT]

#Q. A processor can support a maximum memory of 4 GB, where the memory is word addressable (a word consists of two bytes). The size of the address bus of the processor is at least ____ bits?

[NAT]

#Q. Consider a memory of size $2K \times 8$-bits. What is the size of decoder needed to access the cells of the memory uniquely?

[NAT]

#Q. If there are m input lines n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of m + n is ______?

2 mins Summary

Topic Memory Hierarchy

Topic Memory Presentation

Happy Learning

THANK - YOU